

Information Aggregation in Dynamic Markets with Strategic Traders

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Abstract

This paper studies information aggregation in dynamic markets with partially informed strategic traders. A natural condition on traded securities and information structures, “separability,” is introduced. If a security is separable under a given information structure, then information about its value always gets aggregated, for any prior distribution over the states of the world: In equilibrium, the market price of the security converges to its expected value conditional on the traders’ pooled information. If the security is non-separable, then there exists a prior such that information does not get aggregated. Special cases satisfying separability include, among others, Arrow-Debreu securities, whose value is equal to one in one state of the world and to zero in all others, and “additive” securities, whose value is equal to the sum of traders’ signals.

Keywords: information aggregation, market efficiency, arbitrage, common priors, agreeing to disagree, prediction markets.

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1 Introduction

The idea that financial markets have the ability to aggregate and reveal dispersed information is an important part of economic thinking. The intuition behind this idea is arbitrage: if the price of a security is wrong, an informed trader will have an incentive to buy or sell this security, thus bringing the price closer to the correct value. This intuition is very compelling when one or more traders are fully informed and know the value of the security. It is also compelling in many cases where each trader is “small” relative to the market and behaves, in essence, non-strategically, thus revealing all his information. But what happens when there is a small number of large, strategic players, and none of them is fully informed about the value of the security? What if one trader has perfect information about one part of a company and another trader has perfect information about the rest of the company? Will the stock price reflect the true value of the company that the traders could estimate by pooling their information? Or is there a chance that the price will be off? What happens when information is dispersed among many agents in the economy and their knowledge structure is more complex?

This paper shows that the answers to these questions depend on a natural condition on traded securities and information structures: “separability.” A security is traded in a dynamic market, in which every trader has infinitely many opportunities to trade. The main result of the paper is that in equilibrium, information about a separable security always gets aggregated: its market price converges in probability to its expected value conditional on the traders’ pooled information, for any prior over the states of the world. In contrast, if the security is non-separable, then there exists a prior such that information does not get aggregated.

In the model, there is a finite number of possible states of the world and one traded security, which takes a particular value in each state. There is also a finite number of players, whose information about the states of the world is represented by Aumann’s (1976) partition structure. The players share a common prior belief about the probability of each state. The security is called separable under the partition structure if, for *every* prior, either (i) the value of the security is the same in all states of the world that have positive probability under the prior or (ii) there exist a trader and two states of the world that occur with positive probability under the prior such that the trader’s expectation of the value of the security conditional on the

information that he observes in one state is not equal to his expectation conditional on the information that he observes in the other state. Arrow-Debreu securities (which have payoff 1 in one state of the world and 0 in all other states of the world), additive securities (for which the value of the security is equal to the sum of traders' signals) and their monotone transformations (e.g., multiplicative securities and call options on additive and multiplicative securities), and securities whose payoff is equal to the minimum, maximum, or any other order statistic of players' signals are all separable. A simple example of a non-separable security is the following. Suppose there are two players, each tossing a coin and observing only his own toss. If both coins turn up heads or both coins turn up tails, the value of the security is 1. Otherwise, the value of the security is -1 . This security is non-separable, because if the coins are fair and the tosses are independent, then both conditions (i) and (ii) above are violated.

There are many ways to model a dynamic market: with or without market makers, with or without noise traders, and so on. In each of them, from the no-trade theorem (Milgrom and Stokey, 1982; Sebenius and Geanakoplos, 1983), we know that if every trader and market maker is fully strategic, shares the same prior, and has no liquidity need for trading, then no profitable trading will take place. There needs to be a "source" of profits, which loses money in expectation, thus stimulating trade and price discovery. In the model used in this paper, this source is an automated market maker. There are no noise traders, and no strategic market makers: the only players are the partially informed traders. Specifically, the models of trading considered in the paper are the market scoring rule of Hanson (2003) and the discounted market scoring rule of Dimitrov and Sami (2008). I will refer to these games collectively as market scoring rule (MSR) games.

Under the market scoring rule, the game starts with a number written on the board—the market maker's initial prediction about the value of the security. The first player can erase that number and write his own prediction, which is observed by all players. Then the second player can further modify the prediction, and so on until the last player, after which the first player can again modify the prediction, and the whole cycle repeats an infinite number of times. The fact that there is an infinite number of trading periods does not mean that the game never ends. Rather, it is a convenient discrete analogue of continuous trading games, with trading open at times $t_0 < t_1 < \dots$ in some bounded time interval. After the game is over, the true value of the security is revealed, and each prediction is evaluated according to a strictly proper

scoring rule s . The payoff of a player from each revision is the difference between the score of his prediction and the score of the previous player’s prediction—in essence, the player “buys out” the previous player’s forecast. The total payoff of a player from the game is the sum of payoffs from all revisions. Players are risk-neutral. The discounted market scoring rule is very similar, except that the total payoff of a player is equal to the discounted sum of payoffs from all revisions, where the payoff from a revision made in period t_k is multiplied by β^k for some $\beta < 1$.

Note three things about MSR games. First, the losses of the market maker are deterministically bounded. Second, each player can always guarantee himself the payoff of zero from a revision by not modifying the number on the board. Third, if each player behaves myopically in each period, the number he will put on the board is his posterior belief about the expected value of the security, given his initial information and the history of revisions up to that point, and thus the “game” turns into the communication process of Geanakoplos and Polemarchakis (1982).

MSR games look very different from a typical trading process in which securities are bought and sold, but in fact the former can be reinterpreted as the latter (see Hanson, 2003; Pennock, 2006; and Chen and Pennock, 2007), with an automated market maker who continuously adjusts the price of the securities as traders buy them from him. Moreover, market scoring rules are increasingly widely used in practice to organize prediction markets (companies like Inkling Markets, Consensus Point, and Xpree, among others, run MSR-based internal prediction markets for a number of large corporations). Finally, the intuition behind the main result is quite general, and, as discussed in Section 5, the techniques used in the proof are likely to apply to a variety of other market microstructures. The main advantage of using MSR games as a modeling device is that it brings the key issue studied in this paper—aggregation of strategic traders’ information—to the forefront. At the same time, it eliminates the need to consider noise traders, strategic market makers, and other aspects of more typical trading games that would divert attention from the key issue.

The main result of the current paper says that if a security is separable, then in the limit, the number written on the board will converge to the expectation of the security’s value conditional on all traders’ pooled information. The intuition behind the proof is as follows. Consider an uninformed outside observer who has the same prior as the informed traders, receives no direct information about the state of the world, and observes all the forecasts made by the traders. Consider the

stochastic process that corresponds to the observer’s vector of posterior beliefs about the likelihoods of the states of the world after each forecast revision. By construction, this process is a bounded martingale, and therefore, by the martingale convergence theorem, converges to some vector-valued random variable Q_∞ . If Q_∞ puts positive weights on two states of the world in which the value of the security is different, then separability implies that there is a player who can, in expectation, make a non-vanishing positive profit by revising the number on the board in any sufficiently late period. This, in turn, can be shown to imply that the player is not maximizing his payoff (because he never actually makes that deviation), which is impossible in equilibrium. Thus, with probability 1, Q_∞ has to put all weight on states in which the value of the security is the same. Since the beliefs have to be on average correct, this is only possible if this value is in fact the correct one with probability 1. Now, if the random variable does put all weight on the states with the correct value of the security, but the number on the board does not converge to the same value, then even the uninformed outside observer could have made a profitable revision in infinitely many periods, and thus any informed player could have made such revisions as well, again contradicting the assumption of profit-maximizing behavior. Therefore, the outsider’s posterior beliefs, in the limit, have to put all weight on the states with the correct value of the security, and the number on the board has to converge to the same value.

The question of information aggregation in markets has attracted the attention of many economists, going back to Hayek (1945). Summarizing this research is far beyond the scope of this paper, and recent surveys of theoretical, empirical, and experimental results on the topic are available elsewhere (see, e.g., chapters on experimental asset markets in Kagel and Roth, 1995 and Plott and Smith, 2008; Wolfers and Zitzewitz, 2004; and Vives, 2008). I discuss below the strands of theoretical literature most directly related to the current paper.

Grossman (1976) formally shows that in a market equilibrium, the resulting price aggregates information dispersed among n -types of informed traders, each of whom gets a “piece of information.” In his model, individual traders are small relative to the market, strategic foundations for players’ behavior are lacking, and the results rely on particular functional forms assumed in the model (e.g., i.i.d. normal errors in signals received by the players; normal prior; etc.). Radner (1979) introduces the general concept of Rational Expectations Equilibrium (REE) and shows that gener-

ically, prices aggregate dispersed information, though his model also lacks strategic foundations.

An important stream of literature offers strategic foundations for the concept of perfect competition with differentially informed agents by studying bidding behavior in single and double auctions (Wilson, 1977; Milgrom, 1981; Pesendorfer and Swinkels, 1997; Reny and Perry, 2006). Information aggregation results in these papers, however, rely heavily on both the assumption that the market is large and on various symmetry and independence assumptions. No such assumptions are made in the current paper. Of course, this stream of literature studies not only the process of price formation, but also allocative efficiency of auctions. Note, however, that in the current paper’s model, the security may be equal to, e.g., the marginal consumer’s valuation for a particular good, if this number can be observed *ex post*, and the public knowledge of the marginal consumer’s valuation is sufficient for allocative efficiency.

Another approach to the analysis of price formation and information aggregation in markets is based on the model of Kyle (1985). In that model, like in the current paper, trading is dynamic, and the informed strategic player can place multiple orders over time. In the continuous version of the model, as time converges to the market closing time, the price of the traded asset converges to its true value, which is known by the informed player, but not by noise traders or market makers. Foster and Viswanathan (1996) and Back, Cao, and Willard (2000) extend the model to the case of multiple, differentially informed strategic traders, and find that in the continuous case, price converges to the expected value of the asset conditional on the traders’ pooled information. In the discrete case with a finite number of trading periods, convergence is approximate. These results are similar in spirit to the convergence result in the current paper, but they rely heavily on very special functional-form assumptions (symmetry, normality, etc.), which allow them to construct explicit formulas for very particular (“linear”) equilibria. No such assumptions are made in the current paper.

Two recent papers have studied equilibrium behavior of traders in MSR games. Chen et al. (2007) consider undiscounted games based on a particular scoring rule—logarithmic (see Section 2). In their model, the security can take one of two different values, and the number of revisions is finite. They find that if traders’ signals are independent conditional on the value of the security, then it is an equilibrium for each trader in each period to behave myopically, i.e., to make the prediction equal to

his posterior belief. They also provide an example of a market in which signals are not conditionally independent and one of the traders has an incentive to behave non-myopically, but do not study information aggregation and convergence properties in infinite dynamic markets when myopic behavior is not optimal. Dimitrov and Sami (2008) also consider games based on the logarithmic scoring rule. In their models, in contrast to Chen et al., traders observe independent signals. Each realization of the vector of signals corresponds to a particular value of the security. The number of trading periods is infinite. Dimitrov and Sami find that in that case, in the MSR game with no discounting, myopic behavior is generically not an equilibrium and, moreover, there is no equilibrium in which all uncertainty is guaranteed to get resolved after a finite number of periods. They then introduce a 2-player 2-signal MSR game with discounting, and prove that in that game, information gets aggregated in the limit, under the additional assumption that the “complementarity bound” of the security is positive. They report that based on their sample configurations, the bound is not always zero, but do not provide any sufficient conditions for it to be positive. In contrast to these two papers, the current paper’s information aggregation results do not rely on independence or conditional independence of signals and instead allow for general information structures with any finite number of players, do not depend on discounting, and provide a sharp characterization of securities for which information always gets aggregated and those for which under some priors, price may not converge to the expected value conditional on the traders’ pooled information.

The rest of the paper is organized as follows. Section 2 presents the formal model. Section 3 states and proves the main result. Section 4 describes classes of separable securities and gives a convenient criterion for separability. Section 5 discusses several important open questions.

2 Model

There are n players, $i = 1, \dots, n$. There is a finite set of states of the world, Ω , and a random variable (“security”) $X : \Omega \rightarrow \mathbb{R}$. Each player i receives information about the true state of the world, $\omega \in \Omega$, according to partition Π_i of Ω (i.e., if the true state of the world is ω , player i observes $\Pi_i(\omega)$). For notational convenience, without loss of generality, I assume that the join (the coarsest common refinement) of partitions $\Pi_1, \dots, \Pi_n$ consists of singleton sets; i.e., for any two states $\omega_1 \neq \omega_2$ there

exists player i such that $\Pi_i(\omega_1) \neq \Pi_i(\omega_2)$. $\Pi = (\Pi_1, \dots, \Pi_n)$ is the *partition structure*. Players have a common prior distribution P over states in Ω .

At time $t_0 = 0$, nature takes a random draw and selects the state of the world, ω^* , according to P . The uninformed market maker makes the initial prediction $y_0 \in \mathbb{R}$ about the value of X (a natural initial value for y_0 is the unconditional expected value of X under P , but it could also be equal to any other real number). At time $t_1 \in (t_0, 1)$, player 1 makes a “revised prediction,” y_1 . At time $t_2 \in (t_1, 1)$, player 2 makes his prediction, y_2 , and so on. At time t_{n+1} , player 1 moves again, and makes his new forecast, y_{n+1} , and the whole process repeats until time $t_\infty = 1$, with players taking turns revising predictions. All predictions y_k are observed by all players. The action space is bounded, but the bounds are wide enough to allow for any prediction consistent with random variable X , i.e., each y_k is a number in an interval $[\underline{y}, \bar{y}]$, where $\underline{y} \leq \min_{\omega \in \Omega} X(\omega) \leq \max_{\omega \in \Omega} X(\omega) \leq \bar{y}$.

At time $t^* > 1$, the true value $x^* = X(\omega^*)$ of the security is revealed. The players’ payoffs are computed according to a market scoring rule that is based on a strictly proper single-period scoring rule s . More formally, a single-period scoring rule is a function $s(y, x^*)$, where x^* is a realization of a random variable and y is a prediction. The scoring rule is proper if for any random variable X , the expectation of s is maximized at $y = E[X]$. It is strictly proper if $y = E[X]$ is the unique prediction maximizing the expected value of s . Examples of strictly proper scoring rules include the quadratic scoring rule ($s(y, x^*) = -(x^* - y)^2$), due to Brier (1950), and, when random variable X is bounded (which, of course, is the case in our setting), the logarithmic scoring rule ($s(y, x^*) = (x^* - a) \ln(y - a) + (b - x^*) \ln(b - y)$, for some $a < \underline{y}$ and $b > \bar{y}$), due to Good (1952). I assume that $s(y, x^*)$ is continuous and bounded on $[\underline{y}, \bar{y}] \times [\min_{\omega \in \Omega} X(\omega), \max_{\omega \in \Omega} X(\omega)]$.

Under the basic market scoring rule (introduced by Hanson, 2003, though the idea of repeatedly using a proper scoring rule to help forecasters aggregate information goes back to McKelvey and Page, 1990), players get multiple chances to make predictions, and are paid for each revision. Specifically, for each revision of the prediction from y_{k-1} to y_k , player i is paid $s(y_k, x^*) - s(y_{k-1}, x^*)$. Of course, this number can turn out to be negative, but each player can guarantee himself a zero payment for a revision by simply setting $y_k = y_{k-1}$, i.e., by not revising the forecast. A slight modification of this payoff rule, introduced by Dimitrov and Sami (2008), is a discounted market scoring rule: it is the same as the basic market scoring rule, except that the payment

for the revision from y_{k-1} to y_k is equal to $\beta^k(s(y_k, x^*) - s(y_{k-1}, x^*))$, $0 < \beta \leq 1$. When $\beta = 1$, this rule coincides with the basic market scoring rule. The total payoff of each player is the sum of all payments for revisions. The players are risk-neutral. The resulting game is denoted $\Gamma(\Omega, \Pi, X, y_0, \underline{y}, \bar{y}, s, \beta)$.

Definition 1 *In a perfect Bayesian equilibrium of game Γ , information gets aggregated if sequence y_k converges in probability to random variable $X(\omega^*)$.*

Since the number of possible states of the world is finite, this definition is equivalent to saying that for any $\epsilon > 0$ and $\delta > 0$, there exists K such that for any $k > K$, for any realization of the nature's draw $\omega^* \in \Omega$, the probability that $|y_k - X(\omega^*)| > \epsilon$ is less than δ .

2.1 Separability

Consider the following example, which is a formal version of the coin-tossing example in the introduction (and is based on an example from Geanakoplos and Polemarchakis, 1982). There are two players, 1 and 2. $\Omega = \{A, B, C, D\}$; $P(\omega) = \frac{1}{4}$ for every $\omega \in \Omega$; $X(A) = X(D) = 1$ and $X(B) = X(C) = -1$. Partitions are $\Pi_1 = \{\{A, B\}, \{C, D\}\}$ and $\Pi_2 = \{\{A, C\}, \{B, D\}\}$. Other parameters of the game are arbitrary.

Consider the following profile of strategies and beliefs of the two players: ($y_k = 0$ for any $k \geq 1$; each player's beliefs put equal probabilities on the two possible observations of the other player, after any history). These strategies and beliefs form a perfect Bayesian equilibrium, in which information does not get aggregated. Even though every player announces his posterior belief truthfully, the other player does not learn anything from the announcement. I call securities for which this situation is possible non-separable. Formally,

Definition 2 *Security X is non-separable under partition structure Π if there exist distribution P on the underlying state space Ω and value $v \in \mathbb{R}$ such that:*

1. $P(\omega)$ is positive on at least one state ω in which $X(\omega) \neq v$;
2. For every player i and every state ω with $P(\omega) > 0$,

$$E[X|\Pi_i(\omega)] = \frac{\sum_{\omega' \in \Pi_i(\omega)} P(\omega')X(\omega')}{\sum_{\omega' \in \Pi_i(\omega)} P(\omega')} = v.$$

Otherwise, security X is separable. Note that non-separable securities are not degenerate—e.g., for any security with payoffs in ϵ -neighborhoods of the ones in the above example, there is a distribution P that would satisfy the conditions in Definition 2, and thus all such securities are non-separable.

3 Main result

The main result of this paper is that information about separable securities always gets aggregated, while for non-separable ones that is not the case.

Theorem 1 *Consider state space Ω , security X , and partition structure Π . If security X is separable under Π , then for any prior distribution P , proper scoring rule s , initial value y_0 , bounds $\underline{y}$ and $\bar{y}$, and discount factor $\beta \in (0, 1]$, information in any perfect Bayesian equilibrium of the corresponding game Γ gets aggregated. If security X is non-separable under Π , then there exists prior P such that for any s , y_0 , $\underline{y}$, $\bar{y}$, and β , there exists a perfect Bayesian equilibrium of the corresponding game Γ in which information does not get aggregated.*

Proof. See Appendix. ■

4 Applications

In light of Theorem 1, it is important to understand how restrictive the separability condition is. This section describes several important classes of separable securities and gives an alternative characterization of the condition.

4.1 Arrow-Debreu securities

First, let us show that any Arrow-Debreu security, i.e., random variable X that is equal to 1 in one state, say ω_1 , and to 0 in all other states, is separable. This is in fact an implication of a more general result (Corollary 1 below), but a direct proof of this statement is simple and instructive.

Indeed, note first that for any Arrow-Debreu security X and any state ω' with $X(\omega') = 0$, there exists at least one player i who knows with certainty that the value of X is 0, i.e., for any $\omega'' \in \Pi_i(\omega')$, $X(\omega'') = 0$. Suppose the security is non-separable

and there exist distribution P and value v satisfying the requirements of Definition 2. P has to be positive on at least two different states (otherwise the two requirements cannot hold together). At least one of these two states, say ω_0 , is such that $X(\omega_0) = 0$. Then, at state ω_0 there is a player who knows that the value of security X is 0, and so v has to be equal to 0. But that is only possible if $P(\omega_1) = 0$, contradicting requirement 1.

Thus, by Theorem 1, information about Arrow-Debreu securities always gets aggregated. While the analysis of markets with multiple securities is beyond the scope of this paper, this result suggests that in complete markets, information always gets aggregated.

4.2 Additive payoffs

Another important class of separable securities are those with additive payoffs. Specifically, suppose each player's information $\Pi_i(\omega)$ can be interpreted as a signal $x_i(\omega) = x_i(\Pi_i(\omega)) \in \mathbb{R}$, so that the value of security $X(\omega)$ is equal to the sum of players' signals, $X(\omega) = \sum_i x_i$. This assumption, of course, also includes a seemingly more general case of $X(\omega)$ being a linear function of x_i 's, such as the average, because x_i 's can be rescaled, or a stochastically monotone function (Nielsen et al., 1990), for the same reason.

Let us prove that X is separable. This result also follows from a more general statement (Corollary 2 below), but the direct proof is more straightforward and, in fact, illustrates the intuition behind the general characterization result in Theorem 2. The direct proof is similar to the proof of Theorem 3 of Nielsen et al. (1990).

Suppose X is not separable. Then there exist prior P and value v satisfying the requirements of Definition 2. Consider the unconditional expectation $E[(X(\omega) - v)^2]$ under P .

On the one hand, from requirement 1 of Definition 2, we know that $E[(X(\omega) - v)^2]$ is positive.

On the other hand, since $E[X(\omega)] = v$ and thus $E[(X(\omega) - v)v] = 0$, we have

$$\begin{aligned}
E[(X(\omega) - v)^2] &= E[(X(\omega) - v)X(\omega)] \\
&= E[(X(\omega) - v) \sum_{i=1, \dots, n} x_i(\omega)] \\
&= \sum_i E[(X(\omega) - v)x_i(\omega)] \\
&= \sum_i \sum_{\omega} P(\omega)(X(\omega) - v)x_i(\omega) \\
&= \sum_i \sum_{\pi \in \Pi_i} \sum_{\omega \in \pi} P(\omega)(X(\omega) - v)x_i(\omega) \\
&= \sum_i \sum_{\pi \in \Pi_i} \sum_{\omega \in \pi} P(\omega)(X(\omega) - v)x_i(\pi) \\
&= \sum_i \sum_{\pi \in \Pi_i} x_i(\pi) \left(\sum_{\omega \in \pi} P(\omega)(X(\omega) - v) \right) = 0.
\end{aligned}$$

The last equality follows from requirement 2 of Definition 2, which implies that for any $\pi \in \Pi_i$, $\sum_{\omega \in \pi} P(\omega)(X(\omega) - v) = 0$. We have arrived at a contradiction.

Thus, information about securities with additive payoffs always gets aggregated, for every distribution of priors, correlation structure of signals, and so on.

4.3 Increasing payoffs

A generalization of securities with additive payoffs are securities with “increasing” payoffs. Specifically, suppose each player’s information $\Pi_i(\omega)$ can be interpreted as a signal $x_i(\omega) = x_i(\Pi_i(\omega)) \in \mathbb{R}$ so that $X(\omega_1) \geq X(\omega_2)$ whenever $x_i(\omega_1) \geq x_i(\omega_2)$ for all players i . If there are two traders in the market, then any such security is separable. To see this, suppose the security is non-separable and take the P and v that verify its non-separability. Take the lowest signal x_1 of player 1 and the corresponding element π_1 of partition Π_1 such that there exist states $\omega \in \pi_1$ with $P(\omega) > 0$ and $X(\omega) \neq v$. Among ω in π_1 for which $P(\omega) > 0$, let ω' be the one with the largest corresponding signal of player 2. Then $X(\omega') > v$ and, by construction and monotonicity, all other states in $\Pi_2(\omega')$ that occur with positive probability under P have associated values of X greater than or equal to v , which contradicts requirement 2 of Definition 2.

With three or more players, it is no longer true that any security with increasing payoffs is separable. Indeed, consider the following example. There are three players,

1, 2, and 3. Each player's signal is equal to 1, 2, or 3. If the sum of signals is less than 6, then the value of security X is -1 . If the sum is greater than 6, the value is 1. If the sum of signals is equal to 6, then the value of security X is -1 if the vector of signals is $(1, 2, 3)$, $(2, 3, 1)$, or $(3, 1, 2)$, 1 if the vector of signals is $(3, 2, 1)$, $(2, 1, 3)$, or $(1, 3, 2)$, and 0 if the vector of signals is $(2, 2, 2)$. It is straightforward to check that value $v = 0$ and prior probability P that places probability $\frac{1}{6}$ on every permutation of $(1, 2, 3)$ and 0 on all other states satisfy the requirements for the non-separability of security X .

4.4 Dual characterization of separable securities

While Definition 2 is very intuitive, there is an alternative way to characterize separable securities, using duality theory.¹ This alternative characterization turns out to be convenient in applications, as Corollaries 1 and 2 illustrate.

Theorem 2 *Security X is separable under partition structure Π if and only if for every $v \in \mathbb{R}$, there exist numbers λ_π corresponding to the elements π of partitions Π_i of all players, such that for every state ω with $X(\omega) \neq v$, the sign of $X(\omega) - v$ is equal to the sign of $\sum_i \lambda_{\Pi_i(\omega)}$, i.e., $(X(\omega) - v)(\sum_i \lambda_{\Pi_i(\omega)}) > 0$.*

Proof. See Appendix. ■

The first corollary of Theorem 2 shows that securities that can be represented as order statistics of traders' signals (minimum, maximum, median, etc.) are separable.

Corollary 1 *Suppose security X can be expressed as an order statistic of players' signals: $X(\omega) = x_{(j)}$ (i.e., the j^{th} lowest signal x), where $x_i = x_i(\Pi_i(\omega))$ is the "signal" observed by player i in the state of the world ω . Then X is separable.*

Proof. Take any $v \in \mathbb{R}$. For every i and ω , set $\lambda_{\Pi_i(\omega)}$ equal to 1 if $x_i(\omega) \geq v$ and to $-\frac{n-j+1}{j-0.5}$ if $x_i(\omega) < v$. Then $X(\omega) > v \Rightarrow x_{(j)} > v \Rightarrow \sum_i \lambda_{\Pi_i(\omega)} \geq (n-j+1) - \frac{n-j+1}{j-0.5}(j-1) > 0$ and $X(\omega) < v \Rightarrow x_{(j)} < v \Rightarrow \sum_i \lambda_{\Pi_i(\omega)} \leq (n-j) - \frac{n-j+1}{j-0.5}j < 0$. ■

The second corollary shows that monotone transformations of additive securities (e.g., call or put options on additive securities) are separable.

¹I am grateful to Yury Makarychev for this result.

Corollary 2 *Suppose security X can be expressed as $X(\omega) = f(\sum_i x_i(\Pi_i(\omega)))$, where $x_i(\Pi_i(\omega))$ is the “signal” observed by player i in state of the world ω and f is a monotone function. Then X is separable.*

Proof. Assume f is increasing (the other case is completely analogous) and continuous (since X takes only a finite number of values, this is w.l.o.g.). Take any $v \in [\min_{\omega \in \Omega} X(\omega), \max_{\omega \in \Omega} X(\omega)]$ (the case where v is outside this interval is trivial). Take any z such that $f(z) = v$. Setting $\lambda_{\Pi_i(\omega)} = x_i(\Pi_i(\omega)) - \frac{z}{n}$ for every player i and state ω and applying Theorem 2 proves the result. ■

5 Conclusion: open questions

This paper leaves several important questions for future research. The first one is whether perfect Bayesian equilibria always exist in the games studied in the paper. A discrete version of discounted ($\beta < 1$) MSR game Γ , in which players are only allowed to pick predictions from a finite set of values, is continuous at infinity, and therefore has a perfect Bayesian equilibrium (Fudenberg and Levine, 1983). In the current paper’s model, however, action spaces are continuous, and so equilibrium existence is an open question, for both discounted and undiscounted MSR games.

Another important question is what happens when a security is non-separable, but the prior is generic. E.g., suppose the security and the partition structure are as in the example in Section 2, but the prior is a small generic perturbation of the one in the example. Then if the players simply announced their posterior beliefs truthfully, as in Geanakoplos and Polemarchakis (1982), information would get aggregated. What happens in the strategic trading game? Does there exist an equilibrium in which information gets aggregated with probability 1? Does there exist an equilibrium in which with positive probability information does not get aggregated, and instead in the limit, players get “stuck” at (or converge to) a prediction and a profile of beliefs under which none of them can make a profitable revision? Are the answers the same for all non-separable securities, scoring rules, and other parameters of the game?

There is also a number of interesting questions that go beyond the current paper’s model. First, the intuition behind the main result of the paper suggests that it is very general and should continue to hold in many other market microstructure models, with liquidity-driven or noise traders, strategic market makers, and so on.

The key ingredients that drive the proof are the facts that (i) the outside observer's belief process is a martingale, which has to converge to a limit random variable and (ii) if the traded security is separable and there is a positive probability that the limit random variable places a positive weight on a state with an incorrect value of the security, then that implies that at least one trader, with positive probability, has infinitely many opportunities to make a profitable deviation, but never does. So the result should continue to hold in other microstructure models in which a trader who believes that the current price of a security is incorrect can make a profit by moving the price closer to the correct value. Nevertheless, details of the trading process may turn out to matter for the results, and so it is important to consider formally other dynamic microstructures and to check in which of them similar conclusions hold.

Second, the model in the current paper assumes that traders are risk neutral. What happens if they are not? What happens if their utility functions are different?

Finally, this paper assumes that traders already possess the information, and the only concern is whether this information will get aggregated in the market. But markets are also often viewed as an incentive mechanism for traders to gather costly information, not just as an aggregation mechanism. What happens when traders can acquire information at a cost? How well is information extracted and aggregated in that case? This paper shows that for separable securities, information gets aggregated regardless of which strictly proper scoring rule is used, and the intuition suggests that the aggregation result would continue to hold for other dynamic market mechanisms. With costly information acquisition, however, different mechanisms and different scoring rules may have different properties, giving rise to mechanism design questions.

A Proofs

Proof of Theorem 1

Let $w = 1, \dots, |\Omega|$ index the states in Ω . Let $r = (r^1, r^2, \dots, r^{|\Omega|})$ be a probability distribution over the states (with $r^w \equiv r(\omega_w)$ denoting the probability of state ω_w) and let z be any real number. Define *instant opportunity* of player i as the highest expected payoff he can receive from making only one change to the forecast if the state is drawn according to distribution r and the current forecast on the board is z . Formally, the instant opportunity of player i given r and z is equal to

$$\sum_{\omega \in \Omega} r(\omega) (s(E_r[X|\Pi_i(\omega)], X(\omega)) - s(z, X(\omega))).$$

We first make an auxiliary observation. Let Δ be the set of probability distributions r such that there are states ω_a and ω_b with $r(\omega_a) > 0$, $r(\omega_b) > 0$, and $X(\omega_a) \neq X(\omega_b)$.

Lemma 1 *If security X is separable, then $\forall r \in \Delta \exists \phi > 0, i \in \{1, 2, \dots, n\}$ such that for any $z \in [\underline{y}, \bar{y}]$, the instant opportunity of player i given r and z is greater than ϕ .*

Proof. Consider separable security X and $r \in \Delta$ such that for any trader i and any $\phi > 0$, there exists $z \in [\underline{y}, \bar{y}]$ such that the instant opportunity of i given r and z is less than ϕ . By continuity of score function s , for every trader i there exists $z_i \in [\underline{y}, \bar{y}]$ such that the instant opportunity of i given r and z_i is equal to zero. Since s is strictly proper, this implies that for any ω with $r(\omega) > 0$, $E_r[X|\Pi_i(\omega)] = z_i$, which in turn implies that every z_i is equal to $E_r[X]$. This contradicts the assumption that security X is separable. ■

Now, let $q_0^w = P(\omega_w)$, i.e., the prior probability of state ω_w . Take a PBE of game Γ and consider the following stochastic process Y in $\mathbb{R}^{|\Omega|+1}$. Y_0 is deterministic and is equal to $(y_0, q_0^1, q_0^2, \dots, q_0^{|\Omega|})$. Then nature draws state ω at random, according to distribution P , and each player i observes $\Pi_i(\omega)$. After that, player 1 plays according to his equilibrium strategy and makes forecast y_1 . Based on this forecast y_1 , the equilibrium strategy of player 1, and the prior P , a Bayesian outside observer, who shares prior P with the traders and observes all forecasts y_k but does not directly observe any information about the realized state ω , can form posterior beliefs about the probability of each state ω_w . Denote this probability by q_1^w . Y_1 is equal to $(y_1, q_1^1, q_1^2, \dots, q_1^{|\Omega|})$. The rest of the process is constructed analogously: $Y_k = (y_k, q_k^1, q_k^2, \dots, q_k^{|\Omega|})$, where y_k

is the forecast made at time t_k and q_k^w is the posterior belief of the Bayesian outside observer about the probability of state ω , given his prior P , equilibrium strategies of players, and their history of forecasts up to and including time t_k .

Of course, $Q = \{(q_k^1, \dots, q_k^{|\Omega|})\}_{k=0,1,\dots}$ is also a stochastic process in $\mathbb{R}^{|\Omega|}$. The key idea of the proof is that this process is a martingale, by the law of iterated expectations. And by the martingale convergence theorem, it has to converge to a random variable, $Q_\infty = (q_\infty^1, \dots, q_\infty^{|\Omega|})$.²

Suppose the statement of Theorem 1 does not hold for this equilibrium. Consider the limit random variable Q_∞ and two possible cases.

Case 1

Suppose there is a positive probability that Q_∞ assigns positive likelihoods to two states ω_a and ω_b with $X(\omega_a) \neq X(\omega_b)$. This implies that there is a vector of posterior probabilities $r = (r^1, \dots, r^{|\Omega|})$ such that $r^a > 0$, $r^b > 0$, and for any $\epsilon > 0$, the probability that Q_∞ is in the ϵ -neighborhood of r is positive. Since Q_k converges to Q_∞ , for any $\epsilon > 0$, there exists time K and $\zeta > 0$ such that for any $k > K$, the probability that Q_k is in the ϵ -neighborhood of r is greater than ζ .

Now, by Lemma 1, for some $\phi > 0$ and player i , the instant opportunity of player i is greater than ϕ given r and any $z \in [\underline{y}, \bar{y}]$. By continuity, this implies that for some $\epsilon > 0$, the instant opportunity of player i is greater than ϕ for any $z \in [\underline{y}, \bar{y}]$ and any vector of probabilities r' in the ϵ -neighborhood of r .

Therefore, for some player i , time t_K , and $\eta > 0$, the expected (over all realizations of stochastic process Q) instant opportunity of player i at any time $t_{n\kappa+i} > t_K$ is greater than η .

Case 2

Suppose now that for every realization of Q_∞ , if positive likelihoods are assigned to states ω_a and ω_b , then $X(\omega_a) = X(\omega_b)$, i.e., in the limit, the outside observer always believes that with probability 1, the value of the security is equal to some x^* , and places zero likelihood on all other possible values. Suppose the true state of the world is ω , which has a positive prior probability of occurring. Then by Bayes' rule, it can only happen with zero probability that the outside observer's posterior beliefs Q_∞ place zero likelihood on the value of the security being equal to $X(\omega)$. (To see this, let H be the set of histories $(y_1, y_2, \dots)$ after which the outside observer

²Since the process is bounded, and thus uniformly integrable, convergence is both with probability 1 and in L^1 . See, e.g., Øksendal (1995, Appendix C) for details.

places zero likelihood on the true state being ω . Then $Prob(\omega|H) = 0$. But then $Prob(H|\omega) = Prob(\omega|H)Prob(H)/Prob(\omega) = 0$.) Hence, for every realization ω of nature's draw, with probability 1, Q_∞ will place likelihood 1 on the value of the security being equal to $X(\omega)$, i.e., in the limit, the outside observer's belief about the value of the security converges to its true value (even though his belief about the state of the world itself does not have to converge to the truth, if there are multiple states in which the security has the same value).

Suppose now that process y_k does not converge in probability to the true value of the security. That is, there exist state ω and numbers $\epsilon > 0$ and $\delta > 0$ such that after state ω is drawn by nature, for any K , there exists $k > K$ such that $Prob(|y_k - X(\omega)| > \epsilon) > \delta$. This, together with the fact that even for the uninformed outside observer the belief about the value of the security converges to the correct one with probability 1, implies that for some player i and $\eta > 0$, for any K , there exists time $t_{n\kappa+i} > t_K$ at which the expected instant opportunity of player i is greater than η .

Crucially, in both Case 1 and Case 2, there exist player i^* and value $\eta^* > 0$ such that there is an infinite number of times $n\kappa + i^*$ in which the expected instant opportunity of player i^* is greater than η^* . Fix i^* and η^* .

Now, let S_k be the expected score of prediction y_k (where the expectation is over all draws of nature and moves by players). The expected payoff to the player who moves in period t_k (it is always the same player) from the forecast revision made in that period is $\beta^k(S_k - S_{k-1})$.

The rest of the proof is split into two parts, depending on the value of parameter β : $\beta < 1$ and $\beta = 1$.

Part “ $\beta < 1$ ”

Take any period t_k . Let Ψ_k be the sum of all players' expected payoffs from the revision made in periods k and later, divided by β^k : $\Psi_k = (S_k - S_{k-1}) + \beta(S_{k+1} - S_k) + \beta^2(S_{k+2} - S_{k+1}) + \dots$. We can make two observations about Ψ_k . First, it is non-negative, because each player can guarantee himself a payoff of zero. Second, for a similar reason, it is greater than or equal to the expected instant opportunity of the player who makes the forecast at time t_k . Consider now $\lim_{K \rightarrow \infty} \sum_{k=1}^K \Psi_k$.

On the one hand, under both Case 1 and Case 2, this limit has to be infinite, because each term Ψ_k is non-negative, and an infinite number of them are greater than η^* .

On the other hand, for any K , $\sum_{k=1}^K \Psi_k =$

$$\begin{aligned}
& (S_1 - S_0) + \beta(S_2 - S_1) + \beta^2(S_3 - S_2) + \dots \\
+ & (S_2 - S_1) + \beta(S_3 - S_2) + \beta^2(S_4 - S_3) + \dots \\
+ & (S_3 - S_2) + \beta(S_4 - S_3) + \beta^2(S_5 - S_4) + \dots \\
+ & \vdots \\
+ & (S_K - S_{K-1}) + \beta(S_{K+1} - S_K) + \beta^2(S_{K+2} - S_{K+1}) + \dots
\end{aligned}$$

$= \sum_{k=0}^{\infty} \beta^k (S_{k+K} - S_k) < \frac{2M}{1-\beta}$, where M is some number greater than the highest possible absolute value of the score function, s , given the possible range of outcomes $[\min_{\omega \in \Omega} X(\omega), \max_{\omega \in \Omega} X(\omega)]$ and the allowed range of predictions $[\underline{y}, \bar{y}]$. Hence, both Cases 1 and 2 are impossible, and so y_k has to converge to the true value of security X .

Part “ $\beta = 1$ ”

Take any player i . His expected payoff is equal to $\sum_{j=1}^{\infty} (S_{i+nj} - S_{i+nj-1})$. In equilibrium, the players' expected payoffs exist and are finite, so the infinite sum has to converge. Therefore, for any $\epsilon > 0$ there exists J such that for any $j > J$, $|\sum_{j'=j}^{\infty} (S_{i+nj'} - S_{i+nj'-1})| < \epsilon$. But in both Case 1 and Case 2, that contradicts the assumption that players are profit-maximizing after any history. To see that, it is enough to consider player i^* and some period t_{nj+i^*} such that the expected instant opportunity of i^* is greater than η^* and $|\sum_{j'=j}^{\infty} (S_{i^*+nj'} - S_{i^*+nj'-1})|$ is less than η^* .

Proof of Theorem 2

Notice first that strengthening requirement 1 in Definition 2 to “ $P(\omega) = 0$ for every state ω such that $X(\omega) = v$ ” results in an equivalent definition. To see this, take a non-separable security X , take P and v satisfying the existing definition, set $P(\omega)$ to zero for all ω such that $X(\omega) = v$, and rescale the probabilities of remaining states so that they add up to one. Requirement 2 is unaffected, so the new definition is also satisfied.

Now, take any state space Ω , partition structure Π of players $i = 1, \dots, n$, and

security X . Take any $v \in \mathbb{R}$. Ignore all states ω with $X(\omega) = v$, and let $w = 1, \dots, W$ index the remaining states. Let $m = 1, \dots, M$ index all elements π of all players' partitions. Construct an $M \times W$ matrix A as follows. If state ω_w is in subset π_m , then the element in row m and column w of the matrix is equal to $X(\omega_w) - v$. Otherwise, it is equal to zero.

By Gordan's Transposition Theorem,³ exactly one of the following two systems of equations and inequalities has a solution:

1. $Ax = 0, x \geq 0, x \neq 0$ (where $x \in \mathbb{R}^W$);
2. $A^T \lambda > 0$ (where $\lambda \in \mathbb{R}^M$).

Thus, either

- for some v , system (1) has a solution, and so the security is non-separable (Take solution x of (1); rescale it so that its elements add up to 1; and use rescaled probabilities as the common prior P . Then $AP = 0$, which implies requirement 2 of the definition of non-separability.)

or

- for every v , system (2) has a solution.

Also, tautologically, every security is either separable or non-separable. Therefore, every security is separable if and only if for every v , system (2) has a solution. That is exactly the statement of Theorem 2.

³<http://eom.springer.de/m/m130240.htm>

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